



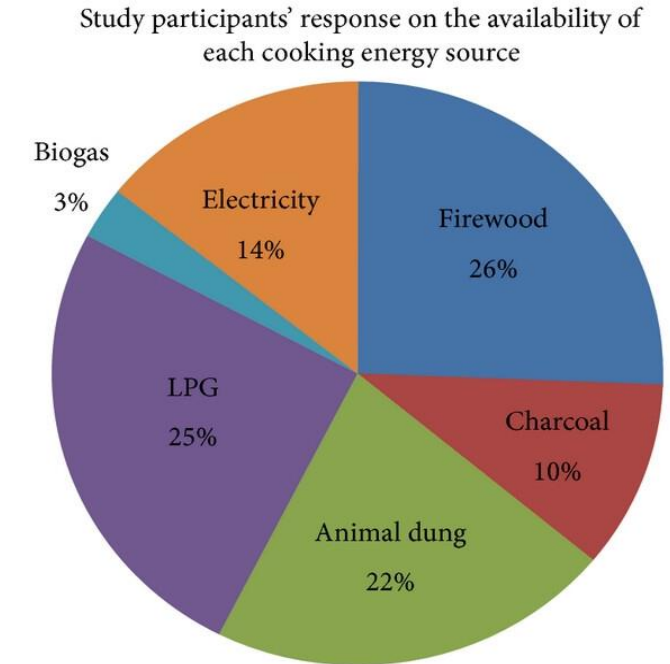
Scenario-Based Modeling of Electricity Demand for Cooking in Ethiopia's Residential Sector Using the OSeMOSYS Model



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Context

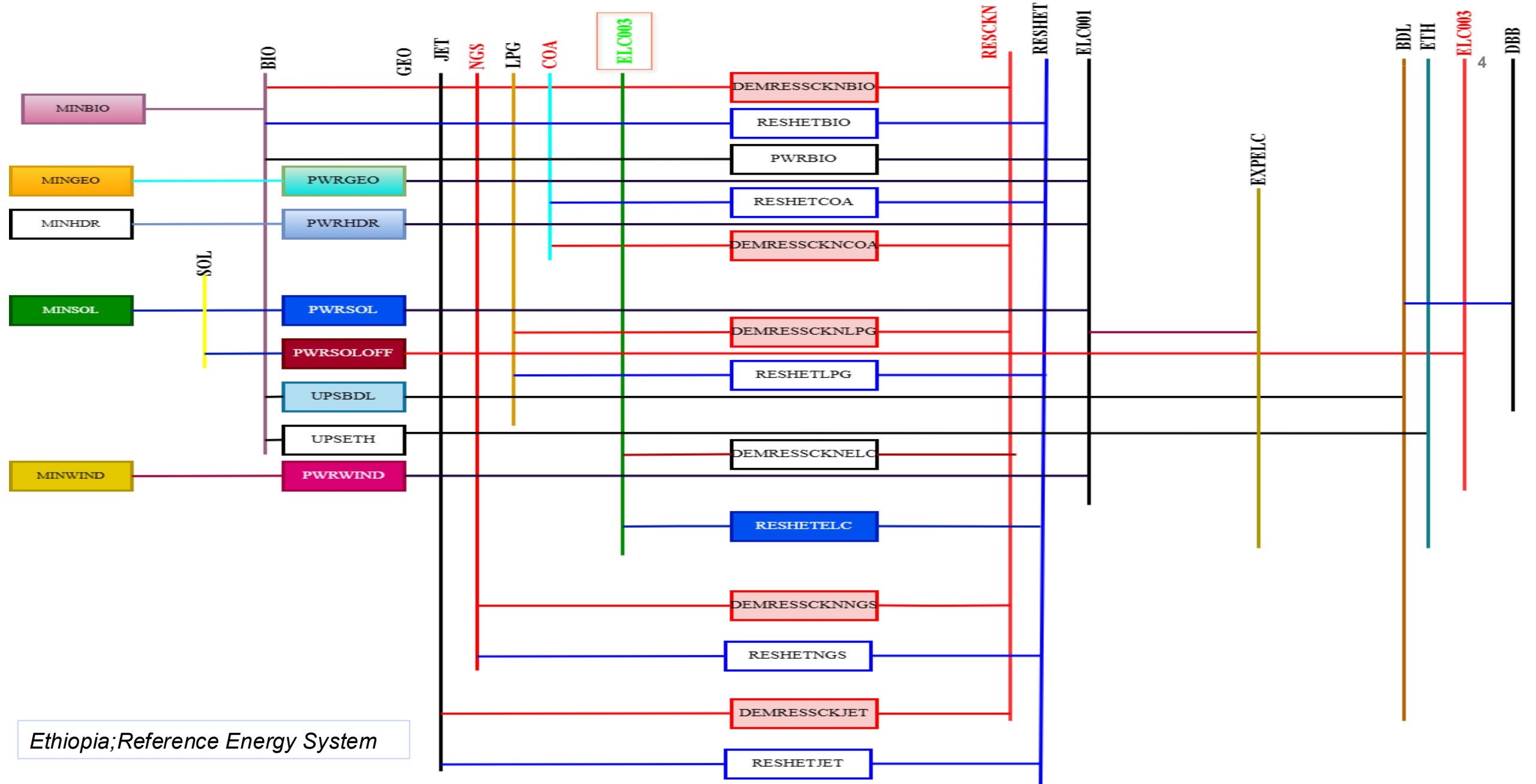
- Promote clean cooking in Ethiopia by improving biomass stove efficiency in the short term and gradually expanding affordable access to electric cooking and LPG in urban and peri-urban areas, supported by electrification and renewable energy expansion to reduce environmental impacts by 2030.



Sources: *WHO & clean cooking studies.*

Research Questions

- *What are the most effective pathways for Ethiopia's residential sector to transition toward clean cooking and electrification?*
- *How can renewable energy expansion sustainably meet future household energy demand?*
- *What environmental and health benefits result from reducing reliance on traditional biomass fuels?*



Context, Challenges, and Main Findings

Main challenge(s)

- *Heavy reliance on traditional biomass for cooking and heating*
- *Limited access to modern energy and reliable electricity in rural areas*
- *Rising residential energy demand with health and environmental impacts*
- *Uncertainty on how fast clean cooking, electrification, and renewables can expand together*
- *The key observation in this scenario is the dominance of wind production*

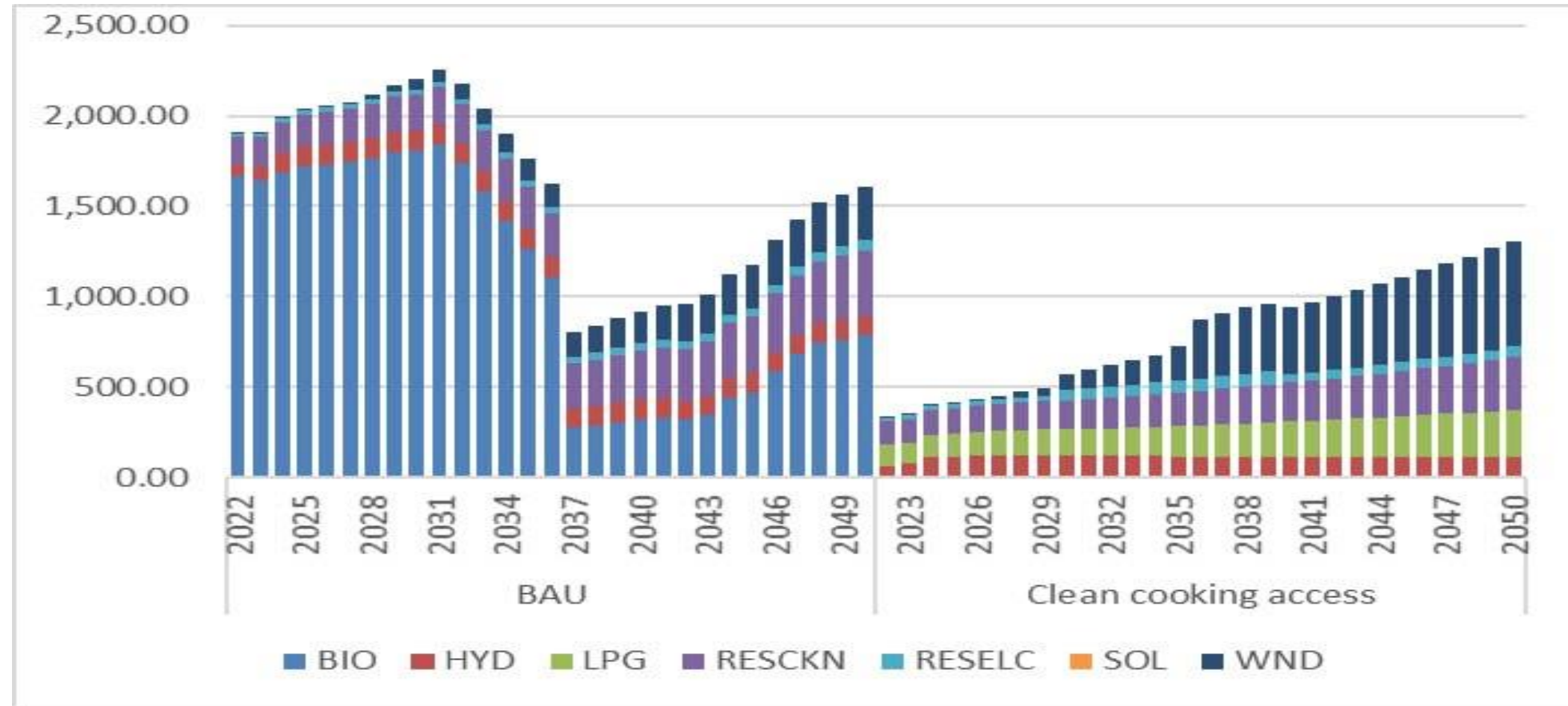
Scenarios

Using the OSeMOSYS Linear Programming Energy System Model the following scenarios were investigated:

Scenario Label	Scenario Description	Key Assumptions
<i>BAU</i>	<i>Continuation of current policies and historical trends in household energy use and electricity expansion, with no additional strong interventions beyond what is already planned.</i>	➤ <i>Current policies and trends continue.</i> ➤ <i>Slow growth in clean cooking adoption.</i> ➤ <i>Electrification follows existing national plans.</i> ➤ <i>Traditional biomass remains dominant in rural areas.</i>
<i>CCA</i>	<i>Accelerated transition from traditional biomass cooking to modern clean cooking options to increase access, reduce indoor air pollution, and lower biomass dependence.</i>	➤ <i>Rapid increase in clean cooking adoption.</i> ➤ <i>Deployment of modern fuels/technologies (e-cooking, LPG, biogas, improved stoves).</i> ➤ <i>Policy and financial support reduce adoption barriers.</i> ➤ <i>Declining share of traditional biomass for cooking.</i>

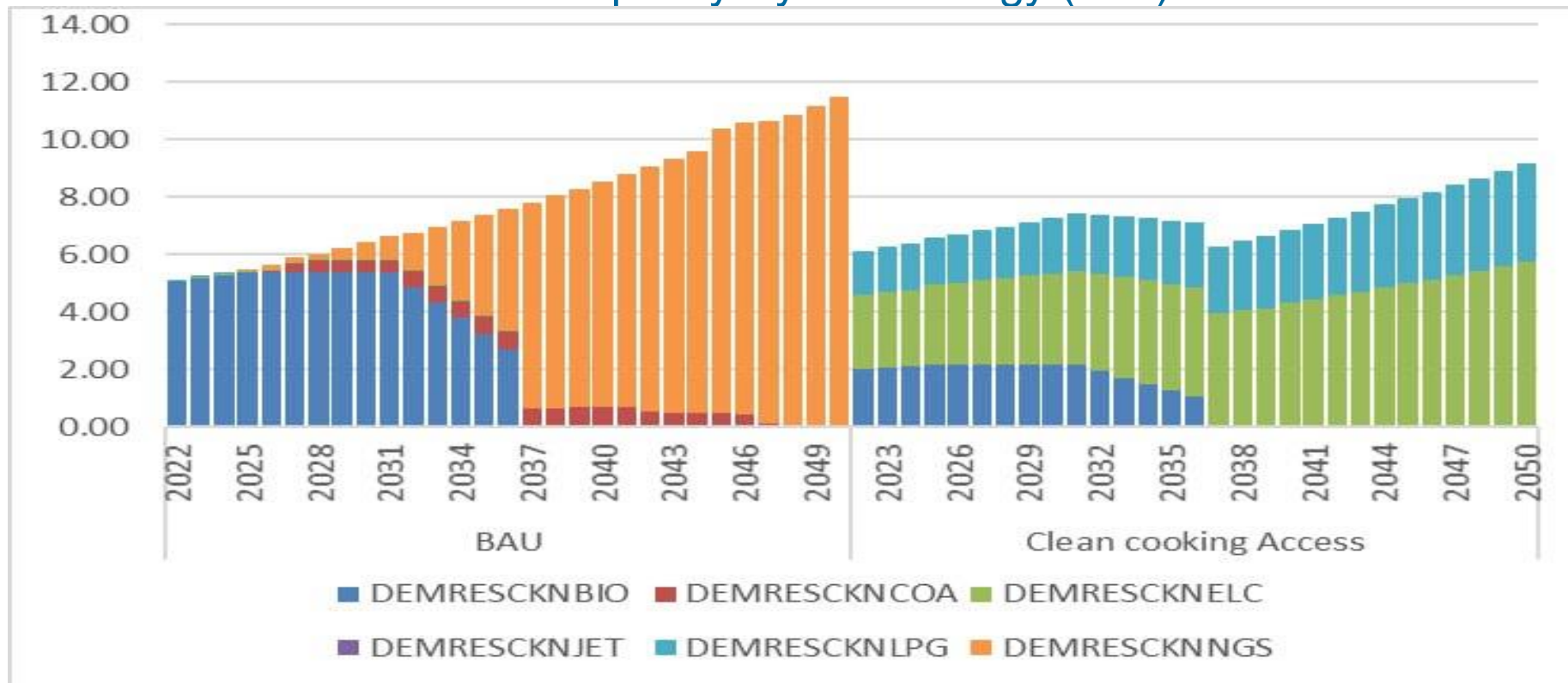
Production by Technology By Mode (PJ)

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- **BAU scenario:** Energy production is dominated by biomass, with limited growth in hydropower and other renewables, indicating continued dependence on traditional technologies.
- **Clean cooking access scenario:** Production shifts toward electricity-based renewables (hydropower, solar, wind) and LPG, with a significant decline in biomass, demonstrating a cleaner and more diversified energy mix.

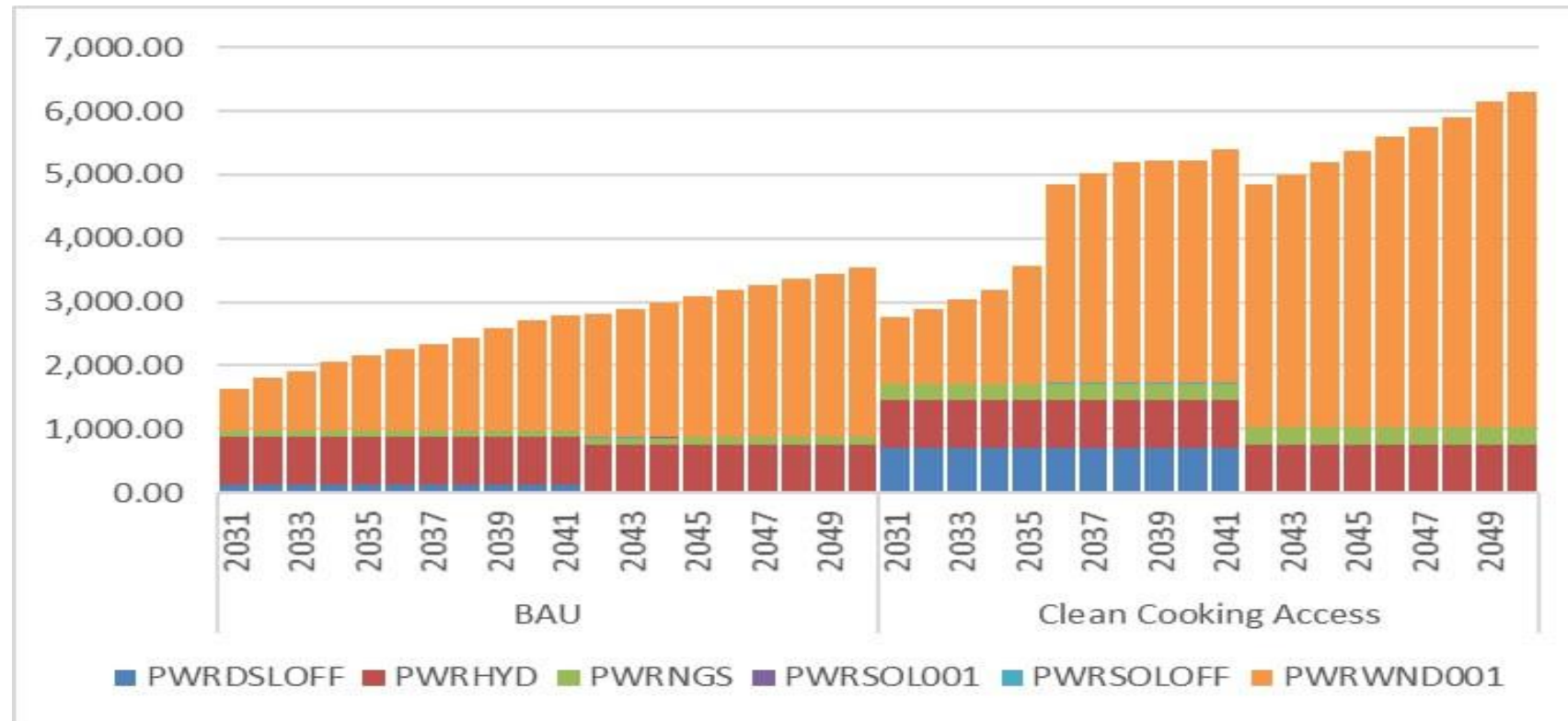
Total Capacity By Technology (GW)



- **BAU scenario:** Total capacity growth is dominated by continued reliance on biomass and gaseous fuels, with minimal uptake of electricity and clean technologies, reinforcing traditional energy use patterns.
- **Clean cooking access scenario:** Total capacity shifts decisively toward electricity and LPG, with a substantial reduction in biomass, reflecting a structural transition to cleaner and more sustainable cooking technologies.

Annualized Investment cost (MUSD/GW)

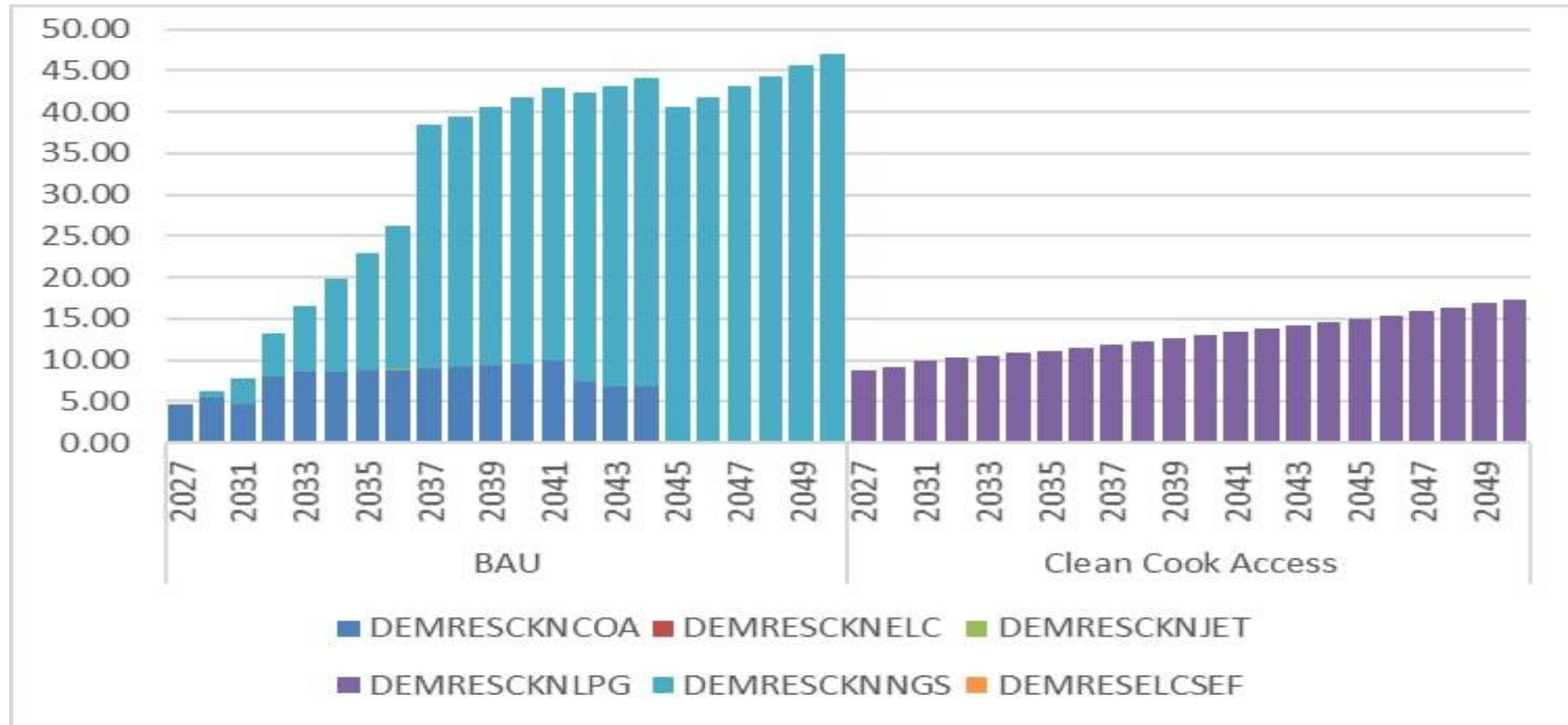
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- **The Clean Cooking Access** scenario's annual investment cost exceeds the BAU due to high capital needs for scaling renewable energy. While hydro remains stable, the scenario invests heavily in wind doubling the BAU projection to reach over 60 units by 2050.
- Most of this spending occurs in the mid-2030s, supporting a shift to electricity and LPG, replacing traditional biomass with capital-intensive energy sources.

Annual Technology Emission (MtonCO2)

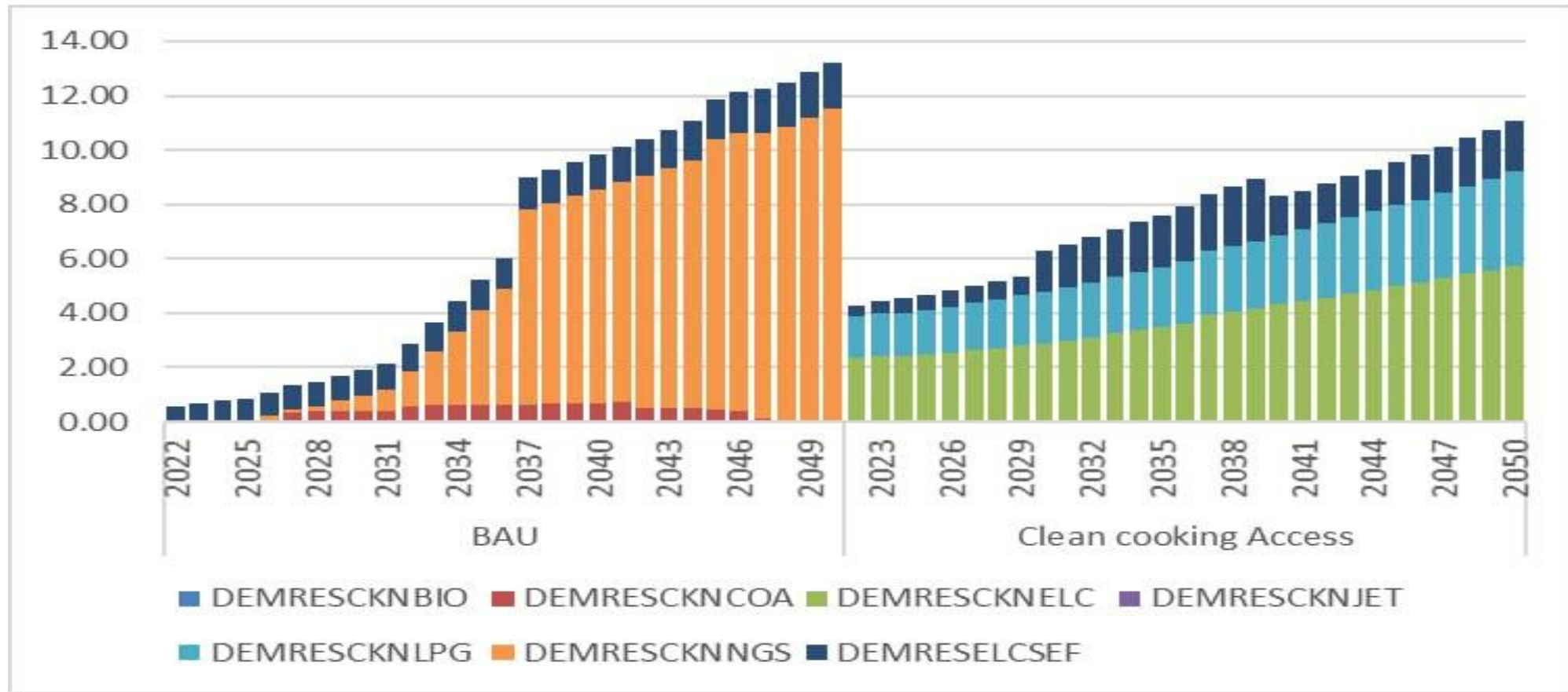
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- The graph shows that Under **BAU**, emissions rise sharply to 2049, driven by wind and gas. In contrast, the Clean Cooking Access scenario keeps emissions lower and steadier, reflecting a shift to cleaner energy and supporting reduction targets.

Accumulated New Capacity (GW)

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The graph shows BAU's capacity growth is dominated by natural gas, highlighting reliance on conventional fuels. In contrast, the Clean Cooking Access scenario diversifies with steady increases in electricity, LPG, and biomass, indicating a shift toward cleaner, modern energy sources and a gradual move away from traditional biomass.

Conclusions and Policy Insights

Main insights

- Clean cooking expansion reduces biomass use and indoor air pollution.
- Renewable-based electrification meets demand with lower fossil dependence.
- Combining both gives the largest health, access, and sustainability benefits.
- lower long-term system costs by reducing fuel imports and inefficient technologies.

Recommended Actions

- Scale up clean cooking technologies with financial support.
- Prioritize renewable power investments.
- Expand grid and off-grid electrification.
- Promote electric cooking in newly electrified areas.
- Gradually reduce traditional biomass reliance.

Conclusions and Policy Insights

Cost Implications

- Requires higher upfront investment in infrastructure, clean cooking technologies, and renewable generation.
- leads to long-term savings from reduced fuel imports, lower health costs, and avoided environmental degradation.
- Requires International climate and development financing can help support the transition.

Future Work

- Improve data on household cooking fuel use and technology performance.
- Include health and environmental impact costs in future model versions.
- Refine regional disaggregation to better capture rural–urban differences.
- Explore policy scenarios such as carbon pricing or fuel subsidy reform.
- Use the model as a capacity-building tool for national energy planning.

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THANK YOU!!